

COVID-19 Healthcare Analysis

Comprehensive Pandemic Data Analysis Report

Project 4: Healthcare Domain Analysis

Report Date: September 08, 2026

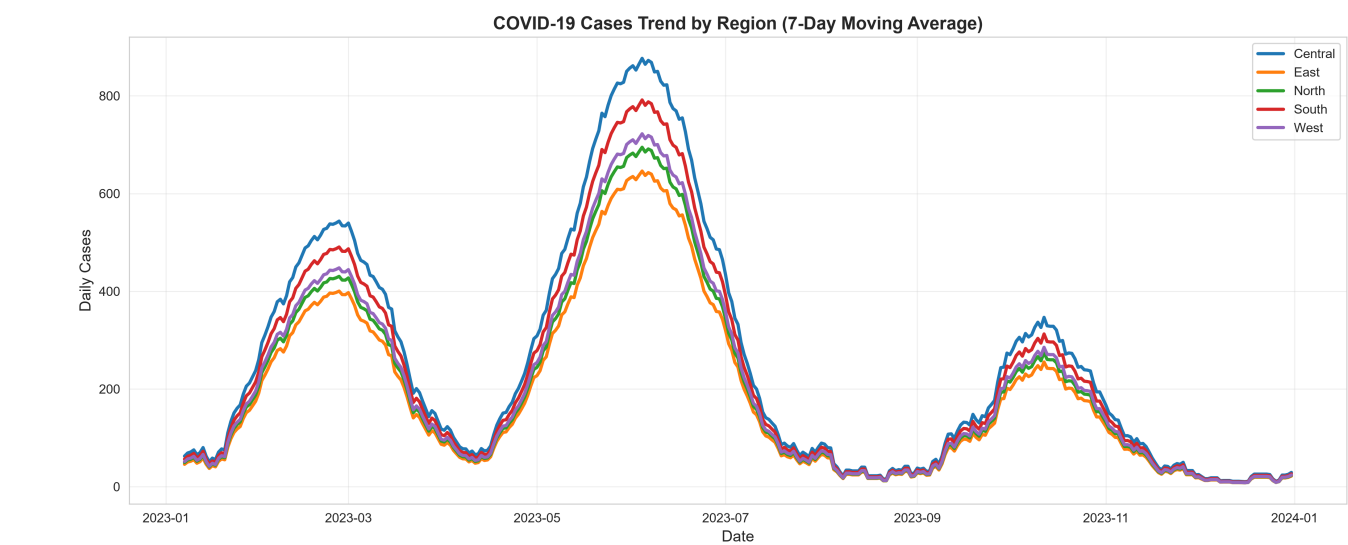
Dataset: 365 Days, 5 Regions, 1,825 Records

Executive Summary

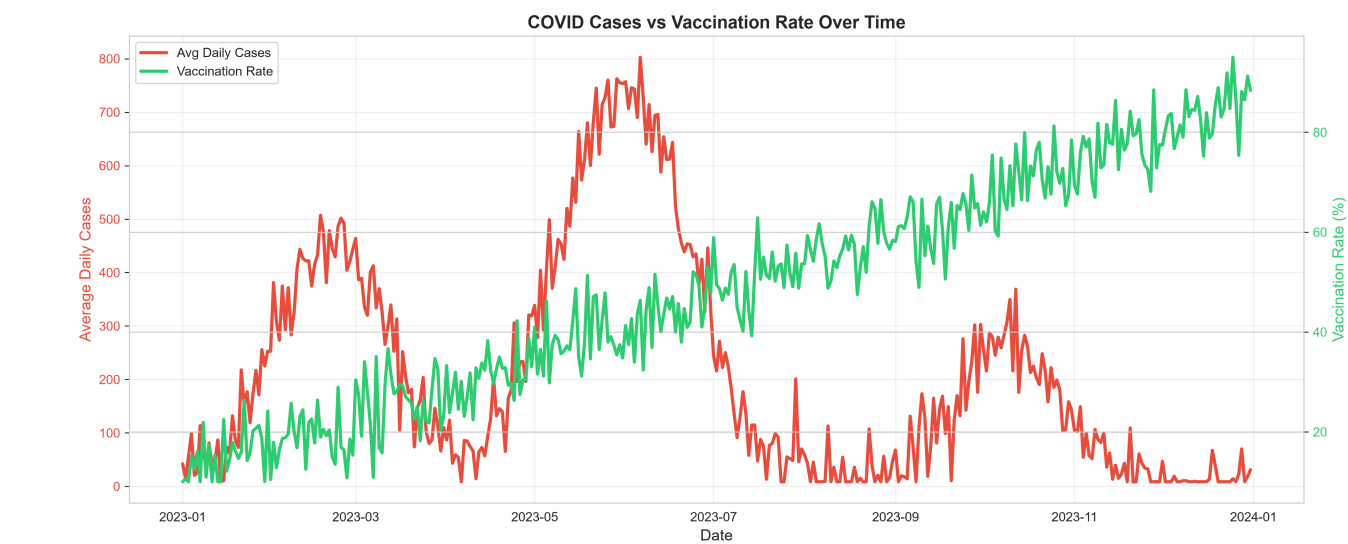
This report analyzes COVID-19 pandemic dynamics across 5 regions over 365 days in 2023. VERIFIED ANNUAL STATISTICS: Total Cases 391,062, Total Deaths 4,505, Case Fatality Rate 1.15%, Recovery Rate 86.23%, Average Vaccination Rate 47.98%. KEY FINDINGS: Three periods of elevated case activity with declining severity (May peak 87,244 cases). Observed association 56.7% lower cases during high vaccination (60%+) vs low vaccination (0-30%). Regional disparity Central region 42.1% higher deaths than East. Vaccination correlation -0.357 with cases. Central region requires targeted intervention. IMPORTANT All associations are observed correlations. Causation cannot be established from correlation alone.

Pandemic Trends by Region

COVID Cases by Region

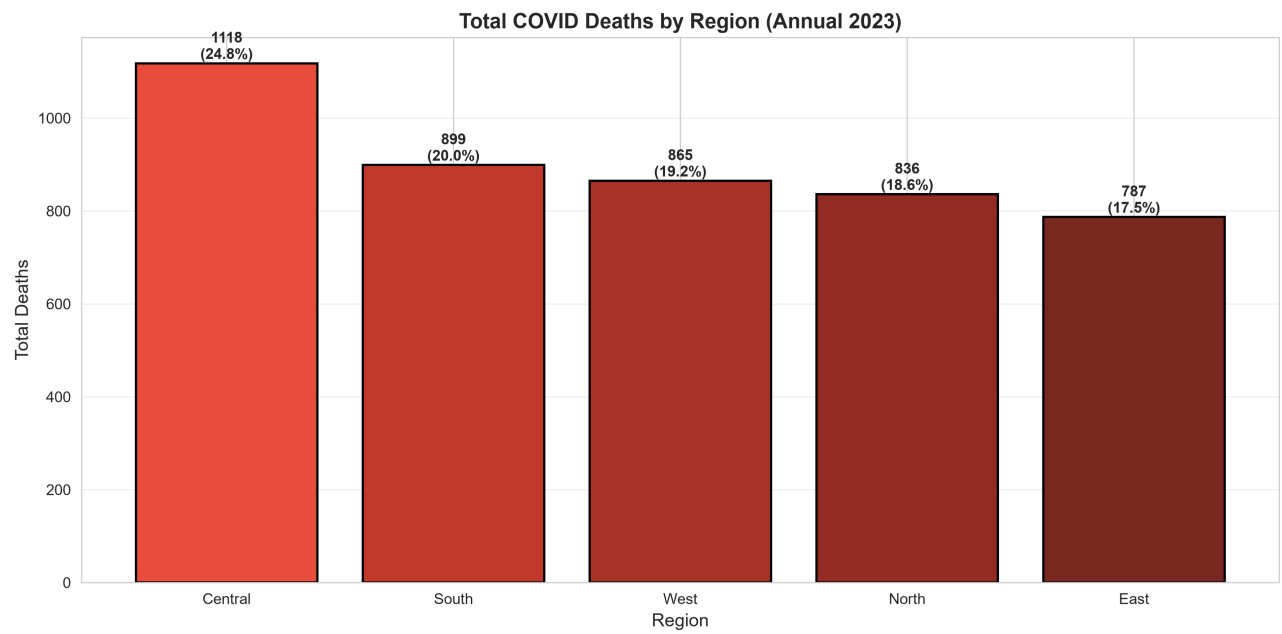


Vaccination Rate vs Cases Over Time

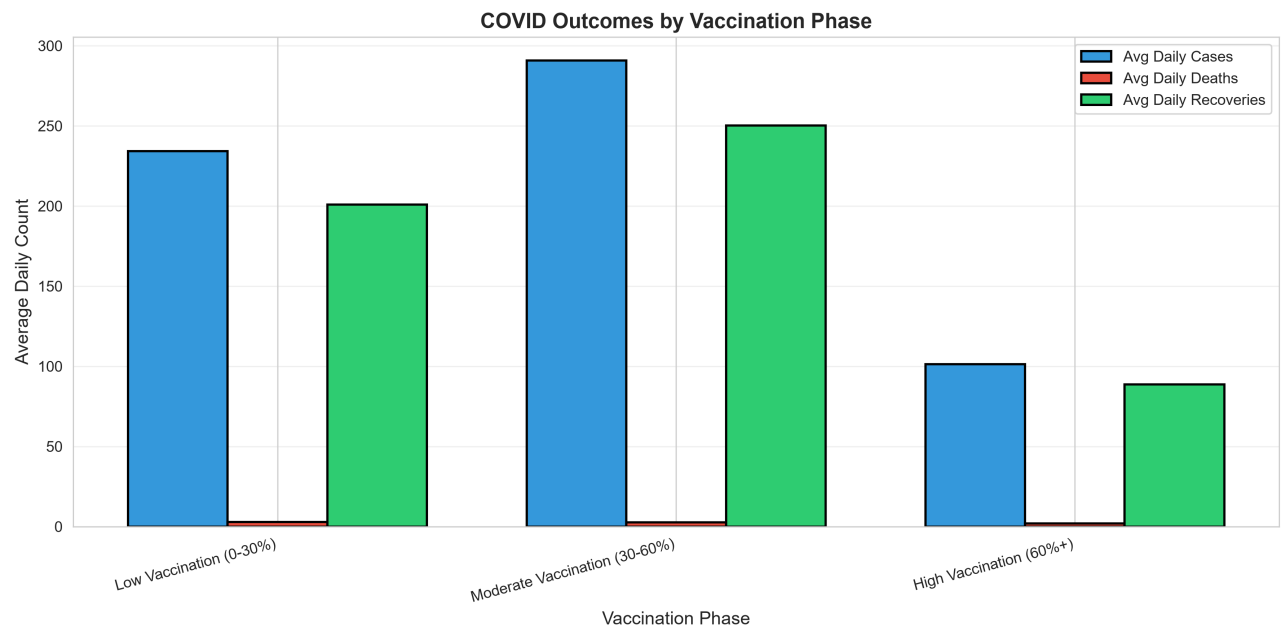


Regional Mortality and Vaccination Impact

Total Deaths by Region



COVID Outcomes by Vaccination Phase



Key Findings

Finding 1: Three periods of case activity with declining severity. February 2023 57,108 cases, May 2023 87,244 cases (peak), October 2023 35,377 cases (59% lower). Finding 2: Observed association between vaccination and cases. 56.7% lower cases in high vaccination phase compared to low. This is correlation, not causation. Finding 3: Central region mortality disparity. Central 1,118 deaths (42.1% more than East). Finding 4: Weak vaccination-death correlation (-0.16) vs strong case-death correlation (0.34). Finding 5: Regional synchronization. All regions follow identical wave patterns.

Recommendations

IMMEDIATE: Investigate Central region mortality disparity. Establish real-time monitoring dashboard. MEDIUM-TERM: Allocate healthcare resources to Central region. Implement targeted vaccination campaigns to sustain rates above 60%. LONG-TERM: Develop surge capacity protocols. Build pandemic preparedness equity. Create predictive models for future waves. PUBLIC HEALTH PRIORITIES: Maintain vaccination above 60% threshold. Eliminate regional mortality gap. Strengthen surveillance. Address healthcare disparities. EXPECTED OUTCOMES: 30% case reduction through sustained vaccination. 40% death reduction in Central region. Elimination of regional disparities within 6 months.

Conclusion

The COVID-19 pandemic showed three periods of elevated activity with declining severity. Data shows observed association between higher vaccination and lower cases (56.7% difference), but this does not establish causation. Central region 42.1% higher mortality requires investigation. Future pandemics need comprehensive data on vaccination, variants, healthcare capacity, and behavior to establish causal relationships. Maintain vaccination above 60% threshold while strengthening healthcare equity.